

Econ 302: Fall 2019 (Eudey)

Fist Exam

Version 2 (Yellow Exam Booklet)

Name\_\_\_\_\_

Discussion Section Instructor\_\_\_\_\_

Discussion Section Number\_\_\_\_\_

The exam has 40 multiple choice questions each worth 1 point.

**Please carefully follow these instructions (*0.5 point is deducted from your total score if it is not completed correctly*):**

- Write your name on your ScanTron
- Write the number of your discussion section under "Special Codes" spaces ABC of your ScanTron form and fill in the bubbles.
- Write the version number of your exam booklet under space D of your ScanTron form and fill in the bubbles.

Example: 3012 (discussion section number 301 + version number 2)

|                      |                                           |                                             |                                             |                                            |
|----------------------|-------------------------------------------|---------------------------------------------|---------------------------------------------|--------------------------------------------|
| Annemarie Schweinert | DIS 303<br>Fr 8:50-9:40<br>(Van Hise 586) | DIS 301<br>Fr 9:55-10:45<br>(Van Hise 590)  | DIS 302<br>Fr 11:00-11:50<br>(Van Hise 209) | DIS 308<br>Fr 1:20-2:10<br>(Sterling 1407) |
| Jack Garton          | DIS 326<br>Fr 8:50-9:40<br>(Van Hise 582) | DIS 325<br>Fr 12:05-12:55<br>(Van Hise 578) | DIS 328<br>Fr 1:20-2:10<br>(Sterling 2323)  | DIS 327<br>Thu 2:25-3:15<br>(Soc Sci 5231) |

No books or notes are allowed during the exam.

1. What color is your exam booklet? (What color is this page?)

- A. White
- B. Yellow
- C. Green
- D. Pink

Answer if Version 2: B

2. The property of diminishing marginal rate of substitution follows from the property that the indifference curve is

- A. Downward-sloping
- ☒ B. Bowed in toward the origin
- C. Rankable
- D. All of these things

$MRS \downarrow$  as  $L \uparrow$  }  $MU(L) \downarrow$   
 "convex" left to right  
 $L \uparrow \Rightarrow MRS \downarrow$



3. If a forecaster uses a log-linear utility function to make a labor market forecast over (for each of) the next twelve months, the forecaster is likely going to

- A. Over-estimate the increase in labor supply in response to a rising real wage
- ☒ B. Under-estimate the increase in labor supply in response to a rising real wage
- C. Over-estimate the decrease in labor supply in response to a rising real wage
- D. Under-estimate the decrease in labor supply in response to a rising real wage

if  $w \uparrow$  in data expect  $N \uparrow$

4. Which effect (income or substitution) always dominates when there is a change in the lump-sum tax?

- ☒ A. Income effect dominates
- B. Substitution effect dominates
- C. The two effects cancel out
- D. We cannot be sure which effect is larger

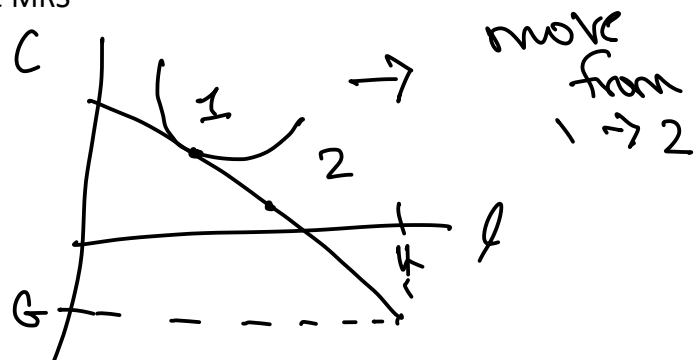
(no substitution effect)

5. If wages remain constant over time, but we observe people working less, what can we say about the  $MRS_{l,c}$ ? (HINT: think about marginal utilities)

- ☒ A. The MRS must have increased
- B. The MRS must have decreased
- C. The MRS must have remained the same
- D. We cannot be sure what happened to the MRS

$N \downarrow, L \uparrow \Rightarrow MU(L) \uparrow$   
 $\uparrow MRS = \frac{MU(L) \uparrow}{MU(C)}$

PDF steeper at 2  
 $\therefore \uparrow MPN = MRS \uparrow$



6. If there is an increase in TFP,  $z$ , which effect (substitution vs income) will dominate for leisure?

- A. The income effect will always dominate, and leisure will increase
- B. The substitution effect will always dominate, and leisure will decrease
- C. The substitution effect will always dominate, and leisure will increase
- ☒ D. We cannot say what the effect on leisure will be without knowing more about preferences

*benefit*  
 *$w(1-t)$  is OC of  $l$*

7. Why is it impossible for a consumer to have  $MRS(l,c) = \frac{MU(l)}{MU(c)} > w(1-t)$ ?

- A. This can only happen if a consumer is on a higher indifference curve than is feasible with the current budget constraint
- B. The consumer could increase utility by increasing consumption since an additional unit of consumption is worth more than the after-tax real wage
- C. The relationship fails to take into account lump sum taxes and government spending which impact a consumer's optimal decision
- ☒ D. The consumer could increase utility by increasing leisure since an additional hour of leisure is worth more than the after-tax real wage

$$\pi = zN - zN = 0$$

8. Suppose a firm originally solves  $\Pi = zN - wN$ . There is a technology shock and  $z$  changes to  $z'$ . What happens to firm profits in equilibrium?

- A. Profits increase
- B. Profits decrease
- ☒ C. Profits are unchanged
- D. We do not know

*how solve for  $w$ ?*  
*so that  $\frac{d\pi}{dN} = 0$*

$$0 = z - w \therefore z = w$$

9. Consider the model from class. According to the model, shocks to preferences cannot be a source for business cycle fluctuations because the model counter-factually predicts

- A. Countercyclical consumption.
- ☒ B. Countercyclical wages.
- C. Pro-cyclical labor hours.
- D. Pro-cyclical technology.

*ch3* } *data:*  
 *$c, w, N, Y, z$*   
*all move same direction*

10. In the general equilibrium model from Chapter 5, the Production Possibility Frontier is bowed because of

- ☒ A. The assumption of diminishing returns to labor.
- B. The assumption that government spending doesn't affect output.
- C. The assumption that technological progress is exogenous.
- D. All of the above.

*$MPN \downarrow$  as  $N \uparrow$  ( $L \downarrow$ )*

11. If firms are perfectly competitive, the wage rate  $w$  will be equal to:

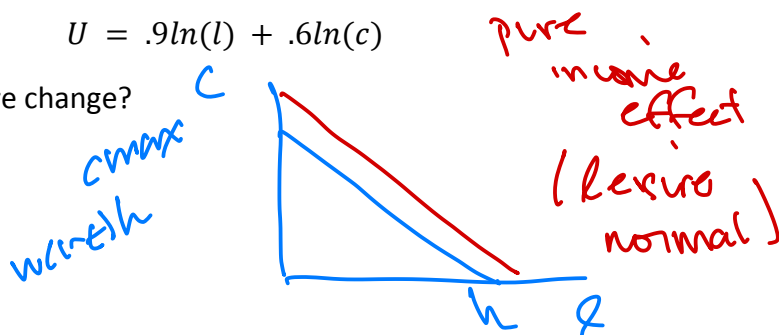
- A. The number of hours worked, divided by total hours
- ☒ B. The marginal productivity of labor
- C. The marginal productivity of capital
- D. We don't have enough information to say what  $w$  will be

12. Suppose Anne drinks a lot of coffee, which increases her daily hours from 16 productive hours in a day to 20 productive hours in a day ( $h$  rises from 16 to 20). Her preferences are captured by

$$U = .9\ln(l) + .6\ln(c)$$

How does her demand for leisure change?

- A. No change
- B. We need more information
- C. Leisure decreases
- ☒ D. Leisure increases



13. In the one-period model, the assumption that everyone would behave the same way under the same circumstance is reflected in

- A. All of the following.
- B. Real wage equaling MPN.
- ☒ C. The use of a representative consumer.
- D. The use of a general equilibrium framework.

Handwritten notes:  $\downarrow$  as  $G \downarrow$ ,  $\uparrow$  as  $N \downarrow$ ,  $\downarrow$  as  $N \downarrow$

14. In our one-period model from class, if government spending decreases, it is most plausible that the model will predict

- A. Increased consumption, increased output, decreased employment, ~~decreased wage rate.~~
- ~~B. Increased taxes, decreased output, increased employment, increased wage rate.~~
- C. Decreased consumption, increased output, increased employment, decreased wage rate.
- ☒ D. Decreased taxes, decreased output, decreased employment, increased wage rate.

Handwritten notes:  $\uparrow MPN = w \uparrow$  as  $N \downarrow$

15. In the general equilibrium model from Chapter 5, an increase in total factor productivity shifts the PPF

- A. Upward, but does not change its slope.
- ☒ B. Upward, and also changes its slope.
- C. Downward, but does not change its slope.
- D. Downward, and also changes its slope.

Handwritten notes:  $\uparrow \text{slope} = MPN \uparrow$  as  $z \uparrow$

16. Which of the following statements best describes trends in US GDP from the mid-1980s onward?

- ch 3
- A. Trend GDP ~~growth is higher~~ and recessions have been more common relative to before 1985
  - B. Trend GDP ~~growth is higher~~ and recessions have been less common relative to before 1985
  - C. Trend GDP growth is lower and recessions have been more common relative to before 1985
  - D. Trend GDP growth is lower and recessions have been less common relative to before 1985

17. Which of the following statements is true of the models that we have seen in Econ 302?

- A. The Social Planner and Representative Household both maximize the same utility function
- B. The Social Planner maximizes efficiency, whereas the Representative Household maximizes utility
- C. The Social Planner and Representative Household have different preferences because the Social Planner cares about everyone in the economy
- D. The Social Planner and the Representative Household both maximize the same utility function if and only if the welfare theorems hold

The next three questions refer to the following economic environment:

$$U = a \ln(C) + b \ln(l)$$

$$C = w(1-t)(h-l) + \pi$$

$$\pi = zN - wN$$

$$Y = zN = C + G$$

18. Which of the following is used in the Social Planner's solution?

- A.  $\frac{b}{a} \cdot \frac{C}{l} = z$
- ~~B.  $\frac{b}{a} \cdot \frac{C}{l} = w(1-t)$~~
- ~~C.  $\frac{a}{b} \cdot \frac{l}{C} = z$~~
- ~~D.  $\frac{a}{b} \cdot \frac{l}{C} = w(1-t)$~~

$MRS = MPN = z$

$\frac{MU(l)}{MU(C)} = \frac{b}{a} \frac{C}{l}$  possible

19. Which of the following is used in the market solution?

- ~~A.  $0 = MRS$~~
- B.  $0 = U'(l)$  ✓
- ~~C.  $0 = \pi = C + G = Y \text{ not } \pi$~~
- ~~D. These are all used to find the market equilibrium~~

need to see ahead of time!!

20. Which of the following is the market solution?

A.  $\frac{b}{a+b} \left[ h - \frac{G}{z} \right] = l$

B.  $\frac{b}{a+b} \left[ h - \frac{G}{w(1-t)} \right] = l$

☒ C.  $\frac{b}{a+b} [h] = l$

D.  $.5[h] = l$

see this in the  
longer ch 5 list for  
(too long for  
this exam)

21. If there is an increase in the real wage rate, has leisure gotten more or less "costly" relative to consumption? (HINT: when I take an extra hour of leisure, what am I giving up?)

A. Leisure has gotten less costly relative to consumption

☒ B. Leisure has gotten more costly relative to consumption

C. There is no change in how costly leisure is relative to consumption

D. The effect is ambiguous

OC of leisure ↑

22. If there is an increase in the proportional tax rate,  $t$ , what can we say happens to our new optimal choice of consumption?

A. Our choice of consumption is higher than it was before the tax increase

☒ B. Our choice of consumption is lower than it was before the tax increase

C. Our choice of consumption is unchanged

D. The effect of the tax on consumption is ambiguous because the tax is proportional

23. Based on the language, theory, and data seen in class, which type of utility function would best be used if one were to be making a long-run forecast for the economy?

A. One that assumes that income and substitution effects do not exist

☒ B. One that assumes that income and substitution effects cancel each other out

C. One that assumes that income effects are larger than (dominate) substitution effects

D. One that assumes that income effects are smaller than (are dominated by) substitution effects

\*\* on old exams we sometimes talk  
about income effects dominating, which is  
wrong

24. Real wages and hours worked are \_\_\_\_\_ correlated over short periods of time and \_\_\_\_\_ correlated over long periods of time.

A. Positively; positively

B. Positively; negatively

C. ~~Negatively; positively~~

D. ~~Negatively; negatively~~

over short  
periods  
uncorrelated  
=

oops  
the data  
were misleading!!  
this is an  
old incorrect  
exam question!

25. Loosely defined, what does the household problem represent? In other words, what are consumers or households choosing?

- A. A consumer maximizes their consumption and leisure allocations
- ☒ B. A consumer chooses the optimal allocation of leisure and consumption within their budget
- C. A consumer maximizes their production of consumption goods subject to technology constraints
- D. None of the above

26. The assumption that leisure is a normal good implies, all other things being equal, that as real wages increase, household will choose to have

- A. Lower labor demand.
- B. Higher labor supply.
- ☒ C. More leisure hours.
- D. More consumption goods.

27. A constant returns to scale production function means

- A.  $F(xK, N) = F(K, xN)$
- ☒ B.  $F(xK, xN) = xF(K, N)$
- C.  $F(xK, xN) = F(K, N)$
- D. All of the above

*if sub effects dominate*

28. What type of relationship between the proportional tax rate  $t$ , and the optimal choice of leisure  $L^*$  is necessary to generate the typical hump-shaped Laffer Curve?

- A. The optimal  $L^*$  must be independent of the tax rate  $t$
- B. The optimal  $L^*$  must be decreasing when  $t$  increases
- ☒ C. The optimal  $L^*$  must be increasing when  $t$  increases
- D. Proportional taxes will always generate a hump-shaped Laffer Curve, regardless of the optimal  $L^*$

$$\text{Rev} = t \cdot \bar{w} \cdot N$$

$\uparrow$                        $\downarrow$

$$(\pi - \tau) \uparrow$$

29. In 2017 the Trump administration passed a significant corporate profits tax cut that affects the non-labor incomes of many US households. Graphically in the chapter 4 model this would be represented by

- ☒ A. A shift out and to the right of the budget constraint; no change in slope
- B. A shift in and to the left of the budget constraint; no change in slope
- C. A shift out and to the right of a steeper budget constraint
- D. A shift in and to the left of a flatter budget constraint

30. Suppose preferences are given by  $U = \ln(C) + \ln(L)$ , and there are no taxes. If we introduce a proportional tax rate  $t$ , what will happen to our optimal choice of leisure,  $L^*$ ?

- A.  $L^*$  will decrease because income effects will dominate substitution effects
- B.  $L^*$  will increase because substitution effects will dominate income effects
- ☒ C.  $L^*$  will be unchanged because income and substitution effects will offset perfectly
- D. The change in  $L^*$  is ambiguous

31. Suppose there is an increase in total factor productivity. How does this change MPK and MPN in the firm's problem if production is Constant Returns to Scale?

- A. MPK and MPN increases

(L↓) N↑ as w↓ } income ↓  
                                Li  
                                normal  
                                good

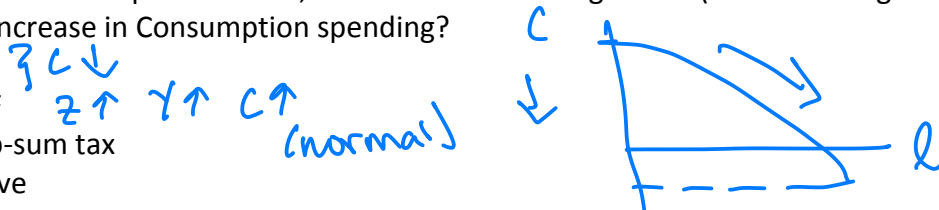
large immediate increase in labor force  
negative shock, you could reasonably infer that that

32. If you saw an economic forecast for a large immediate increase in labor force participation rates in response to a current negative shock, you could reasonably infer that that the forecaster had assumed that

- A. Income effects would dominate (be larger than) substitution effects in the short run
- B. Income effects would be equal to substitution effects in the short run
- C. Income effects would be dominated by (be smaller than) substitution effects in the short run
- D. There weren't either income or substitution effects from the negative shock

33. According to our Chapter 5 model, which of the following would (all other things being equal) lead to an increase in Consumption spending?

- ☒ A. A steeper IC 3 ↓
- ☐ B. A steeper PPF 2 ↑
- ☒ C. A bigger lump-sum tax
- ☒ D. All of the above



34. Assume government cuts payroll taxes and it leads to an increase in tax revenue collections. We would know from this that

- A. Substitution effects dominate income effects *~ Latter Curve*  
B. Income effects dominate substitution effects  
C. Markets are efficient  
D. None of these things can be known with certainty from the information supplied

## ✓ Laffer Curve

35. In the partial equilibrium model from Chapter 4, if dividend income ( $\pi$ ) increases, the following does *not* happen:

- A. The consumer chooses to consume more leisure.
- B. The consumer chooses to consume more consumption goods.
- C. The budget constraint shifts to the right.
- D. The substitution effect exceeds the income effect.

creases, the

A Answer key canvas!!

36. Military spending, environmental protection, education, immunizations, roadwork, and rule of law are all obvious areas of public goods provision where government can correct a private market failure. The language and theory of the class tells us that
- Government involvement in these market can be welfare-improving
  - Government involvement in these markets will likely not bring us to the efficient equilibrium
  - Real-world governments will likely use inefficient means to pay for these services
  - All of these things.
37. Which of the following statements are true about the Marginal Rate of Substitution between leisure and consumption ( $MRS_{l,c}$ )?
- At the consumer's optimal choice of consumption and leisure, the MRS must equal the (negative) slope of their budget constraint
  - The  $MRS_{l,c}$  represents the consumer's willingness to substitute leisure for consumption
  - The  $MRS_{l,c}$  is equal to the (negative) slope of the indifference curve
  - All of the above are true
38. Suppose there is an increase in government spending  $G$ . After the increase, will GDP be greater if we fund the spending with a lump-sum tax  $T$ , or a proportional tax with rate  $t$ ?
- GDP will be greater if we fund the spending with a lump-sum tax
  - GDP will be greater if we fund the spending with a proportional tax
  - GDP will be the same regardless of how we fund the spending
  - We cannot say which case GDP will be larger in without knowing the representative consumer's utility function
39. Which of the following statements is NOT consistent with the data seen in class?
- Americans on average work more than do people with similar incomes in other countries
  - Paid maternity leave policies seem to keep more women in the labor force after having children
  - US population growth has been trending downward for decades and is now around 0.6% per year
  - The long economic expansion since the end of the Great Recession has driven US unemployment rates to zero.
40. Consider the one-period model from class. A decrease in technology would always be associated with
- An increase in GDP.
  - An increase in employment.
  - A decrease in consumption.
  - No change in wage rate.

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